

ATTN: Steven Posnack - Principal Deputy Assistant Secretary for Technology Policy - ASTP/ONC

U.S. Department of Health and Human Services

Submitted by: MAHA Institute

Date: February 2026

Re: HHS Health Sector AI RFI

Dear Mr. Posnack,

Thank you for the opportunity to provide comments on accelerating the adoption and use of AI in clinical care. The innovation of AI has the potential to be transformative to the health of Americans, however with most innovations in healthcare they do not fail because of lack of idea and effort — innovation fails when infrastructure and incentives are misaligned, and the application is not inclusive of the individual.

Meaningful improvement in health and health care requires coordinated governance, interoperable systems, data centered around the person and trust across stakeholders.

Artificial intelligence has the potential to reduce administrative burden, improve timeliness of insight, strengthen monitoring capabilities, and support better clinical decision-making.

AI will not scale in the absence of interoperable data, consistent governance standards, clear regulatory expectations, and payment models aligned with longitudinal value to the individual.

The recommendations included in this submission reflect practical experience navigating complex health journeys, IT implementation, multi-jurisdiction collaboration, and clinical workflow realities. They are focused on the federal policy levers that can create durable conditions for responsible AI adoption.

I appreciate the Department's leadership in seeking broad input and welcome continued engagement as HHS advances its AI strategy.

Respectfully,

Debra Sheldon

Vice President

MAHA Institute

RESPONSE TO “HHS HEALTH SECTOR AI RFI”

Docket: HHS Health Sector AI RFI

Submitted to: Office of the Deputy Secretary and ASTP/ONC

Department of Health and Human Services

Introduction

The enhancement of health and the potential for individuals to be the executive of their health through artificial intelligence is no longer a prospective claim; it is a demonstrable today, in part.

The primary impediment to widespread adoption is not a deficit in model capabilities, but rather a lack of coordinated federal direction on AI infrastructure investments for health should be allocated.

The integration of AI within healthcare delivery remains significantly hindered by individuals not having a centralized easily accessible data sources for consumers of care. In the absence of cohesive federal intervention across regulatory, financial, and research domains, AI deployment will remain inconsistent and limited to constrained, low-risk applications.

The Department of Health and Human Services (HHS) possesses both the requisite authority and the operational mechanisms to effectively dismantle these structural impediments while creating structure to provide adequate protections for patient privacy. The subsequent recommendations propose concrete, actionable policy levers for consideration.

Structural Barriers to Adoption: Fragmented Longitudinal Data Environments

Effective AI necessitates structured, machine-readable, longitudinal data spanning diverse care settings. Despite documented progress under 45 CFR Part 170 (Health IT Certification) and 45 CFR Part 171 (Information Blocking), operational implementation remains highly variable. Access to quality bulk data is inconsistent, semantic normalization is unevenly applied, and standardized audit capabilities across systems are lacking.

Organizations endeavoring to implement cross-institutional AI solutions are compelled to navigate redundant governance structures, inconsistent API implementations, and variable access controls. This fragmentation substantially decelerates innovation and elevates operational costs.

AI scalability is contingent upon interoperability being operationalized, enforceable, and subject to clear measurement.

Regulatory Ambiguity for Non-Device AI. A substantial class of AI tools—including, but not limited to, risk stratification engines, clinical documentation augmentation, workflow optimization platforms, and surveillance triggers—does not satisfy the established criteria for medical device classification, yet directly impacts the delivery of care.

Healthcare systems currently lack definitive federal guidance pertaining to audit requirements, drift monitoring protocols, documentation expectations, and reproducibility standards for these particular tools. This absence of predictable regulatory guardrails prolongs legal review processes and inhibits procurement timelines.

Reimbursement Misalignment. Existing fee-for-service payment models do not systematically incentivize burden reduction, efficiency gains in productivity, or preventative interventions. Furthermore, even within value-based and risk-bearing payment arrangements, the infrastructure investments essential for supporting AI—such as a longitudinal health record accessible by patients, data quality measurement systems, federated query capabilities, and comprehensive monitoring systems—lack explicit recognition.

Liability and Indemnification Uncertainty. Healthcare systems exhibit pervasive caution in deploying AI owing to an unclear demarcation of responsibility between the technology developers and the organizations implementing the solutions. Negotiations concerning indemnification frequently extend procurement cycles beyond a twelve-month horizon.

I. Regulatory Priorities

Despite documented progress under 45 CFR Part 170 (Health IT Certification) and 45 CFR Part 171 (Information Blocking), operational implementation remains highly variable. Access to quality bulk data is inconsistent, semantic normalization is unevenly applied, and standardized audit capabilities across systems are lacking. Additionally, **[add something here about the Privacy Rule]**

To address these regulatory challenges, HHS should adopt a regulatory posture defined by clarity, proportionality, and enforceability.

A. Clarify Governance Expectations Under 45 CFR Part 170. HHS should promulgate interpretive guidance mandating that AI-enabled certified health IT systems must support the following capabilities:

- Model version control and comprehensive documentation
- Tracking of dataset provenance
- Audit logging functionality consistent with the requirements of 45 CFR 170.315(d)
- Reproducible data access when utilizing APIs as defined under 45 CFR 170.315(g)(10)
- Continuous monitoring of model performance and drift
- A consent to opt in/out of state, regional and national networks must be explicit and centralized to reduce friction to the individual
- Health information exchanges need to be reshaped into tools of consent, real world data sources with consumer mediated controls that allow for both privacy protections to be transparent and honored by all actors in the interoperability rule whole allowing the citizen scientist access to aggregated information

These expectations should be applied particularly where AI systems traverse organizational or jurisdictional boundaries.

Predictable governance standards serve to mitigate institutional risk and reduce friction in the procurement process.

- B. Enforce Information Blocking Provisions.** Pursuant to 45 CFR Part 171, HHS must unequivocally state that technical throttling, the imposition of excessive access fees, or proprietary export restrictions that impede access to AI-ready bulk data may constitute interference.

Effective AI deployment is dependent upon access to reliable, timely, and machine-readable electronic health information. Consistent and visible enforcement remains paramount.

- C. Recognize Federated Analytics Architectures.** HHS should formally acknowledge distributed and federated analytics models as compliant with the HIPAA Privacy Rule provisions (45 CFR 164.502 and 164.514), provided that appropriate safeguards are rigorously implemented.

Federated architectures facilitate cross-state and multi-system AI deployment while effectively preserving local data control, supporting role-based access permissions, ensuring auditability, and maintaining small-cell suppression. This model simultaneously reduces privacy risk while significantly expanding analytic capacity.

Federal recognition would furnish crucial assurance to state and regional stakeholders.

A compounding problem for Regulatory Priorities is the lack of regulation for non-device tools. A substantial class of AI tools—including, but not limited to, risk stratification engines, clinical documentation augmentation, workflow optimization platforms, and surveillance triggers—does not satisfy the established criteria for medical device classification, yet directly impacts the delivery of patient care.

Healthcare systems currently lack definitive federal guidance pertaining to audit requirements, drift monitoring protocols, documentation expectations, and reproducibility standards for these particular tools. This absence of predictable regulatory guardrails prolongs legal review processes and inhibits procurement timelines.

II. Reimbursement Modernization

Existing fee-for-service payment models do not systematically incentivize burden reduction, efficiency gains in productivity, or preventative interventions. Furthermore, even within value-based and risk-bearing payment arrangements, the infrastructure investments essential for supporting AI—such as data quality measurement systems, federated query capabilities, and comprehensive monitoring systems—lack explicit financial recognition.

The acceleration of AI adoption is directly contingent upon payment models that recognize both infrastructure investment and longitudinal value creation.

- A. Incorporate AI Infrastructure into Value-Based Arrangements.** Under 42 CFR Part 425 (Medicare Shared Savings Program) and 42 CFR Part 422 (Medicare Advantage), the Centers for Medicare & Medicaid Services (CMS) should permit the inclusion of AI-enabling infrastructure investments—including data normalization efforts, quality measurement cycles, and monitoring systems—within shared savings calculations.

Center for Medicare and Medicaid Innovation (CMMI) demonstrations authorized under 42 U.S.C. § 1315a should explicitly pilot and test AI-enabled care models that holistically integrate analytics, workflow redesign, and governance oversight.

Pilot Innovation Add-On Pathways. CMS should investigate temporary reimbursement mechanisms conceptually analogous to 42 CFR 412.87 for validated AI tools that demonstrably achieve measurable improvements in timeliness, cost reduction, or burden mitigation.

Infrastructure investment must temporally precede the generation of insights. Payment policy should accurately reflect this essential sequence.

III. Research Priorities

HHS should strategically prioritize research in the following critical domains:

- Explainable AI within authentic, auditable, real-world clinical workflows
- Federated multi-jurisdiction analytics methodologies
- Techniques for bias detection in longitudinal datasets
- AI-driven methodologies for data quality improvement
- Human-centered implementation science
- Identify data networks that provide real time, real world data sources that go beyond Acute Care/EHR (Hospital) sources of data

While current literature documents gains in predictive performance and documentation efficiency, empirical evidence regarding system-wide cost redistribution, workforce impact, and long-term savings remains constrained. Federal research investment should be directed toward closing this critical knowledge gap.

Responses to Specific Questions

In addition to responses to the general requests for information above regarding AI regulation, reimbursement, and research & development, we are including responses to the specific questions included in the RFI.

IV. Accountability Frameworks for Non-Device AI

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

- A. Private Sector Innovation.
- B. Adoption and Use in Clinical Care: AI has yielded demonstrable improvements in areas such as imaging triage, ambient documentation, predictive risk stratification within value-based care models, and revenue cycle optimization. These successful implementations share common attributes: seamless integration into existing clinical workflows, access to normalized longitudinal data, and clearly defined monitoring frameworks.

Conversely, AI has underperformed in scenarios where deployment occurred without adequate semantic normalization, lacked integration into core clinical systems, or was implemented without transparent governance controls.

The most promising applications include distributed surveillance systems, longitudinal cross-system risk scoring, AI-assisted methodologies for data quality enhancement, and workflow-integrated clinical augmentation tools.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

HHS should reevaluate the use of health information exchanges as tools for AI. HHS should convene a multistate demonstration of transforming these entities from provider tools to consumer centric, real world data, real world evidence tools of CDC, FDA and NIH. A transformation rooted in Health Data Trust models, that convene states in a coordinated, industry standardizing governance framework must prevail. This transformation would provide create a privacy protecting governance model with an auditable source of truth and create a clear framework for granular consenting to occur with less friction and expanded access to the individual and aid at restoring the public trust in data that is accessible to CDC, FDA, NIH, CMS and beyond.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

Healthcare systems exhibit pervasive caution in deploying AI owing to an unclear demarcation of responsibility between the technology developers and the organizations implementing the solutions. Negotiations concerning indemnification frequently extend procurement cycles beyond a twelve-month horizon. Non-device AI necessitates clear definition of governance responsibilities, specifically:

- The allocation of legal responsibility between the technology developer and the deploying organization.
- Standardized documentation requirements for decisions informed by AI.
- Defined obligations for continuous drift monitoring.
- Transparency requirements for training data provenance and established performance metrics.

HHS should convene cross-agency coordination (ASTP/ONC, CMS, FDA, OCR) to develop standardized governance templates that emphasize auditability, formal approval workflows, mandatory override documentation, and monitoring expectations.

Clear frameworks will reduce variability in negotiation and expedite responsible deployment.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Adoption decisions typically involve a diverse group of stakeholders, including Chief Information Officers (CIOs), Chief Medical Information Officers (CMIOs), compliance officers, Chief Financial Officers (CFOs), legal counsel, and medical executive committees.

The principal impediments encountered include protracted procurement cycles, complex indemnification negotiations, fragmented governance committee structures, and inherent integration constraints within legacy systems.

Federal clarification regarding governance standards and interoperability expectations that are consumer centric would substantially mitigate friction.

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

Enhanced interoperability is essential for expanding AI's utility in population risk management, syndromic surveillance, quality benchmarking, and cross-state analytics.

HHS must continue to prioritize the enforcement of FHIR bulk data access provisions under 45 CFR 170.315(g)(10), structured semantic normalization standards, AI-ready metadata frameworks, and the establishment of measurable data quality benchmarks.

Interoperability must be regarded as essential operational infrastructure, not merely aspirational policy.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Patients expect services characterized by timeliness, personalization, and reduced administrative burden. They simultaneously expect transparency, fairness, and the preservation of necessary clinician oversight.

Human-in-the-loop safeguards, explainability features, and auditable governance structures must be intrinsically embedded into all AI deployment requirements.

Trust must be deliberately engineered; it cannot be presumed.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

Non-device AI necessitates clear definition of governance responsibilities, specifically:

- The allocation of legal responsibility between the technology developer and the deploying organization.
- Standardized documentation requirements for decisions informed by AI.
- Defined obligations for continuous drift monitoring.
- Transparency requirements for training data provenance and established performance metrics.

HHS should convene cross-agency coordination (ASTP/ONC, CMS, FDA, , NIH, CDC, OCR) to develop standardized governance templates that emphasize auditability, formal approval workflows, mandatory override documentation, and monitoring expectations.

Clear frameworks will reduce variability in negotiation and expedite responsible deployment.

V. Evaluation and Continuous Monitoring

AI oversight must adhere to a lifecycle-based methodology.

Pre-deployment evaluation should encompass representativeness analysis, rigorous calibration testing, subpopulation performance assessment, and robustness testing under scenarios of incomplete data.

Post-deployment monitoring must include:

- Continuous detection of model drift
- Performance dashboards for key metrics
- Tracking of clinical overrides
- Scorecards for timeliness and completeness
- Cross-site benchmarking comparisons

HHS can support these necessary processes through cooperative agreements, the establishment of shared benchmarking infrastructure, and the implementation of standardized reporting expectations.

VI. Observed Performance in Clinical Settings

AI has yielded demonstrable improvements in areas such as imaging triage, ambient documentation, predictive risk stratification within value-based care models, and revenue cycle optimization. These successful implementations share common attributes: seamless integration into existing clinical workflows, access to normalized longitudinal data, and clearly defined monitoring frameworks.

Conversely, AI has underperformed in scenarios where deployment occurred without adequate semantic normalization, lacked integration into core clinical systems, or was implemented without transparent governance controls.

The most promising applications include the application to individual management of health if access is truly enabled by the interoperability rule, distributed public health application, longitudinal cross-system risk scoring, AI-assisted methodologies for data quality enhancement, and workflow-integrated clinical augmentation tools.

VII. Organizational Adoption Dynamics

Adoption decisions typically involve a diverse group of stakeholders, including Chief Information Officers (CIOs), Chief Medical Information Officers (CMIOs), compliance officers, Chief Financial Officers (CFOs), legal counsel, and medical executive committees.

The principal impediments encountered include protracted procurement cycles, complex indemnification negotiations, fragmented governance committee structures, and inherent integration constraints within legacy systems.

Federal clarification regarding governance standards and interoperability expectations would substantially mitigate administrative friction.

VIII. Interoperability as a Strategic Enabler

Enhanced interoperability is essential for expanding AI's utility in population risk management, syndromic analysis quality benchmarking, and cross-state analytics.

HHS must continue to prioritize the enforcement of FHIR bulk data access provisions under 45 CFR 170.315(g)(10), structured semantic normalization standards, AI-ready metadata frameworks, and the establishment of measurable data quality benchmarks.

Interoperability must be regarded as essential operational infrastructure, not merely aspirational policy.

IX. Patient and Caregiver Considerations

Patients expect services characterized by timeliness, personalization, and reduced administrative burden. They simultaneously expect transparency, fairness, and the preservation of necessary clinician oversight. Human-in-the-loop safeguards, explainability features, and auditable governance structures must be intrinsically embedded into all AI deployment requirements.

Trust must be deliberately engineered; it cannot be presumed.

X. Research Priorities

HHS should strategically prioritize research in the following critical domains:

- Explainable AI within authentic, real-world clinical workflows
- Federated multi-jurisdiction analytics methodologies
- Techniques for bias detection in longitudinal datasets
- AI-driven methodologies for data quality improvement
- Human-centered implementation science

While current literature documents gains in predictive performance and documentation efficiency, empirical evidence regarding system-wide cost redistribution, workforce impact, and long-term savings remains constrained. Federal research investment should be directed toward closing this critical knowledge gap.

Conclusion

The successful integration of AI into clinical care will accelerate only upon the condition that HHS effectively aligns regulatory clarity, enforceable interoperability standards, reimbursement modernization, and comprehensive lifecycle governance frameworks.

The Department possesses the requisite tools to establish a national framework that enables distributed analytics, maintains local data control, enforces auditability, and aligns financial incentives with the generation of longitudinal value.

In the absence of coordinated federal action, the adoption of AI will remain fragmented. With it, AI can be solidified as durable infrastructure for modern clinical care.

Appendix A: Detailed Responses to HHS Health Sector AI RFI

Comprehensive Analysis of Key Considerations for AI in Healthcare (Questions 1–10)

1. Biggest Barriers to AI Adoption and Scaling in the Health Sector

The primary obstacles preventing the widespread, scaled deployment of artificial intelligence in healthcare are multifaceted and interconnected:

- **Fragmented and Inaccessible Data:** The lack of standardized, high-quality, and easily accessible patient data across disparate systems (e.g., EHRs, imaging archives, claims databases) severely limits the ability to train, validate, and generalize robust AI models. Data silos and inconsistent semantic tagging are persistent issues.
- **Regulatory Ambiguity for Non-Device AI:** While the FDA provides a clear pathway for AI designated as a medical device, there is significant uncertainty regarding the governance, oversight, and liability for clinical, operational, and administrative AI tools that do not meet the definition of a device (e.g., risk stratification tools, documentation assistance, workflow optimizers). This ambiguity stifles innovation and adoption.
- **Misaligned Reimbursement and Economic Models:** Current fee-for-service payment models and established coding structures often fail to adequately recognize or incentivize the use of AI tools, particularly those that drive efficiency, prevention, or improved population health outcomes rather than just procedural volume. A clear, consistent reimbursement pathway is absent.
- **Liability and Accountability Uncertainty:** A lack of clarity regarding liability allocation among developers, implementers, and clinical users in the event of an AI-driven error (e.g., misdiagnosis, care pathway error) creates substantial legal risk and impedes organizational willingness to deploy new technologies. Establishing clear indemnification and accountability frameworks is critical.

2. Necessary Regulatory and Payment Model Changes to Support AI

HHS must take coordinated action across regulatory and payment structures to create an environment conducive to safe and effective AI adoption:

- **Clarify Governance under 45 CFR Part 170 (Health IT Certification):** HHS, through the Office of the National Coordinator for Health Information Technology (ONC), must clearly define the certification criteria and scope of governance for non-device AI tools, ensuring they meet minimum standards for safety, transparency, and data provenance.
- **Strengthen Enforcement of 45 CFR Part 171 (Information Blocking):** Aggressive enforcement of information blocking regulations is essential to unlock the data necessary for model development and validation. This includes focusing on barriers related to API access and the cost-prohibitive nature of data extraction.
- **Recognize and Address Federated Models under HIPAA (45 CFR 164.502 and 164.514):** Policies must explicitly recognize and facilitate federated learning environments, which allow models to be trained across distributed, protected data sets without the need for centralized transfer of Protected Health Information (PHI), ensuring compliance with minimum necessary use principles.
- **Modernize Value-Based Reimbursement (42 CFR Parts 425 and 422):** Payment models for Accountable Care Organizations (ACOs) and Medicare Advantage (MA) plans must be updated to explicitly recognize and incentivize the use of AI that demonstrably improves efficiency, quality metrics, and cost management.

- **Pilot AI-Specific Add-On Payments (Consistent with 42 CFR 412.87):** CMS should establish pilots for new technology add-on payments (NTAPs) specifically tailored to high-value AI solutions to facilitate early adoption and dissemination, similar to mechanisms used for novel medical devices.
- **Expand Center for Medicare and Medicaid Innovation (CMMI) Demonstrations (42 U.S.C. § 1315a):** CMMI should prioritize and scale demonstrations focused on evaluating the cost-effectiveness and clinical impact of AI, particularly in areas like chronic disease management, population health, and reduction of administrative burden.

3. Novel Legal and Policy Issues Requiring Immediate HHS Attention

AI introduces significant novel issues requiring coordinated federal guidance:

- **Liability Allocation and Indemnification:** The most pressing issue is defining clear roles and responsibilities when an error occurs. HHS needs to convene stakeholders to develop model language for contracts and indemnification agreements between AI developers and healthcare providers.
- **Documentation Standards for AI Use:** Clear standards are needed for documenting the use of an AI tool, including the model version, the input data used, the date of last retraining/validation, and the degree to which a clinician overrode the AI recommendation. This is crucial for malpractice defense and quality reporting.
- **Model Drift Accountability:** Policies must address how providers and developers monitor and are held accountable for model performance degradation over time (model drift) due to shifts in patient population, coding changes, or clinical practice patterns.
- **HHS Action Required:** HHS should proactively **issue standardized governance templates and risk assessment frameworks** for AI deployment. Furthermore, it must **convene cross-agency coordination** (including FDA, ONC, CMS, and Office for Civil Rights) to ensure consistent messaging and policy implementation.

4. Effective Methods for Evaluating AI Performance and Safety

Robust evaluation must span the entire lifecycle of the AI model:

- **Pre-Deployment Evaluation:** Rigorous, multi-site testing is required, focusing on **bias detection** (e.g., performance parity across different demographic groups, socioeconomic status, and care settings) and **calibration testing** (ensuring the model's predicted probability aligns with observed outcomes).
- **Post-Deployment Monitoring:** Continuous, real-world monitoring is essential, including tracking for **model drift** (performance degradation) and systematic tracking of **override events** (instances where a clinician rejects or modifies an AI recommendation) to identify potential safety issues or clinical misalignment.
- **Support Mechanisms:** HHS should fund and support this evaluation through **cooperative agreements** for multi-site validation studies and by establishing **benchmarking programs** that provide standardized, de-identified datasets for external model testing and comparison.

5. Role of Accreditation, Certification, and Standards Bodies

Voluntary and collaborative mechanisms can accelerate trust and adoption:

- **Encourage Voluntary AI Accreditation:** HHS should encourage the development of voluntary, third-party AI accreditation programs (e.g., through The Joint Commission or NCQA) that are aligned with federal audit standards for security, equity, and performance documentation.
- **Create Regulatory Testing Sandboxes:** Establish secure, controlled environments ("sandboxes") where developers and providers can test nascent AI tools in a real-world setting under temporary regulatory waivers, allowing for rapid iteration and evidence generation.
- **Develop Model Reporting Templates:** Mandate or strongly encourage the use of standardized reporting templates (e.g., adapted from the Model Card or Data Card concept) to clearly document a model's intended use, performance limitations, training data characteristics, and known bias testing results.
- **Establish Transparency Registries:** Create a centralized, public or semi-public registry for reporting essential metadata on deployed clinical and operational AI models, enhancing transparency for providers, researchers, and patients.

6. Current Performance Experience and Most Promising Future Tools

The health sector has seen varying degrees of success with current AI deployment:

- **Areas of Strong Performance:** AI has shown powerful, validated performance in areas such as **medical imaging** analysis (e.g., radiology, pathology), **clinical documentation** assistance (e.g., ambient listening, charting suggestions), and **predictive modeling** for operational needs (e.g., forecasting patient flow, resource utilization).
- **Areas of Weak Performance and Integration Challenges:** Performance suffers significantly where **integration into existing clinical workflows** is cumbersome, and where **explainability** is absent, leading to a lack of clinician trust and poor adoption rates.
- **Most Promising Future Tools:** The highest future impact is anticipated from **federated surveillance and public health analytics** (allowing models to track disease outbreaks or resource needs without centralizing PHI) and **longitudinal risk scoring** that integrates clinical, claims, and social determinants of health data to proactively manage high-risk patients.

7. Key Organizational Decision Makers and Adoption Hurdles

Adoption requires navigating a complex internal landscape:

- **Key Decision Makers:** Adoption decisions are often a committee-based process involving the **Chief Information Officer (CIO)**, **Chief Medical Information Officer (CMIO)**, **Compliance Officer**, **Chief Financial Officer (CFO)**, and **Legal Counsel**.
- **Primary Adoption Hurdles:**
 - **Lengthy Procurement Cycles:** The time and effort required to move from evaluation to contracting is often prohibitive.
 - **Indemnification Negotiations:** Protracted legal battles over liability and risk sharing between developer and provider.
 - **EHR Integration Barriers:** Technical complexity, vendor-specific limitations, and the cost associated with deep integration into electronic health record systems remain the single largest technical barrier.

8. Interoperability Priorities for AI Development and Deployment

Enhanced, standardized data flow is non-negotiable for AI success:

- **Enforce FHIR Bulk Data Access (45 CFR 170.315(g)(10)):** ONC must prioritize the strict enforcement of the FHIR-based bulk data access requirement, enabling large-scale, efficient data extraction necessary for training and deploying AI models across patient populations.
- **Advance Semantic Normalization and Benchmarking Tools:** Investment and standardization efforts are needed to harmonize disparate clinical terminologies (e.g., LOINC, SNOMED) and administrative codes, enabling models trained on one data set to be reliably applied to another. HHS should support the creation of public-good tools for semantic mapping.

9. Patient Concerns and Needs Regarding AI in Healthcare

Patient perspectives must guide the ethical implementation of AI:

- **Desire for Efficiency and Personalization:** Patients generally welcome AI tools that promise to increase the speed of diagnosis, reduce administrative friction, and deliver more personalized treatment recommendations.
- **Concern about Bias and Opacity:** The chief concerns relate to the potential for AI to perpetuate or amplify existing health inequities (bias) and the lack of understanding about how decisions are made (opacity or the "black box" problem).
- **Required Reinforcements:** Policy must emphasize **radical transparency** regarding when and how AI influences care and mandate robust **human oversight and review** mechanisms, ensuring that the ultimate decision-making authority rests with the human clinician.

10. Critical Research Priorities for Public Investment

Federal research funding should be targeted to address foundational and implementation challenges:

- **Explainable AI (XAI):** Research focused on developing reliable, clinically meaningful methods for explaining AI outputs to both clinicians and patients, moving beyond simple feature importance scores.
- **Federated Analytics and Privacy-Preserving AI:** Investment in technologies like federated learning, homomorphic encryption, and differential privacy to enable model training across sensitive, distributed data sources without violating privacy.
- **Bias Detection, Mitigation, and Health Equity:** Dedicated research on systematic methods for proactively identifying, quantifying, and mitigating algorithmic bias across diverse clinical populations and care settings.
- **Implementation Science:** Studies focused on the socio-technical aspects of AI deployment, including human factors, workflow redesign, and the impact of AI on clinical teams and patient safety culture.
- **Cost-Transfer Analysis:** Rigorous economic research to understand where the costs and benefits of AI are realized (e.g., transfer of cost from the payer to the provider, or from the hospital to the patient) to inform equitable reimbursement policy.